Average and Variance

Problem 791

Denote the average of k numbers $x_1, \ldots, x_k$ by $\bar{x} = \frac{1}{k}\sum_i x_i$. Their variance is defined as $\frac{1}{k}\sum_i (x_i - \bar{x})^2$.

Let $S(n)$ be the sum of all quadruples of integers (a, b, c, d) satisfying $1 \le a \le b \le c \le d \le n$ such that their average is exactly twice their variance.

For $n = 5$, there are 5 such quadruples, namely: $(1,1,1,3), (1,1,3,3), (1,2,3,4), (1,3,4,4), (2,2,3,5)$.

Hence $S(5) = 48$. You are also given $S(10^3) = 37048340$.

Find $S(10^8)$. Give your answer modulo 433494437.